

Chapter

Nervous System and Sense Organs

1. The reflex action is controlled through MAIN 2021

- (a) peripheral nervous system
- (b) autonomic nervous system
- (c) central nervous system
- (d) None of the above

2. The antiseptic present in tears: SQP 2025

- (a) Iodine
- (b) Hydrogen peroxide
- (c) Lysozyme
- (d) Lysosome

**3. The two cerebral hemispheres are joined by a sheet of fibres called:
SQP 2025**

- (a) Pons
- (b) Corpus luteum
- (c) Hypothalamus
- (d) Corpus callosum

4. Rahul inserted a hairpin into his right ear to remove ear wax. He felt a sudden sharp pain with loss of hearing. This was due to: SQP 2025

- (a) Rupture of eardrum
- (b) Rupture of vestibule

(c) Rupture of cornea

(d) Rupture of pinna

5. Assertion (A) : Rods and Cones are photoreceptors in the sclera of eyeball.

Reason (R) : Rods are sensitive to dim light. **SQP 2025**

(a) Assertion is True and Reason is False.

(b) Assertion is False and Reason is True.

(c) Both Assertion and Reason are True.

(d) Both Assertion and Reason are False.

6. Adjusting the focal length of the eye lens to view objects at different distances is done by: MAIN 2024

(a) Cornea

(b) Iris

(c) Ciliary outgrips

(d) Sclera

7. The ear ossicle that transports sound vibrations to the inner ear:

MAIN 2024

(a) Stapes

(b) Mallets

(c) Incas

(d) Cochlea

8. The lower part of brain which contains reflex centre. COMP 2021

9. Pinna COMP 2021

10. Eardrum SQP 2023

11. Eustachian tube MAIN 2022

12. The two hemispheres of the cerebrum are connected by SQP 2023

- (a) corpus callosum
- (b) corpus luteum
- (c) cerebellum
- (d) pons

13. The insulating sheath covering the axon is called COMP 2022

- (a) neurilemma
- (b) plasmalemma
- (c) dura mater
- (d) pia mater

14. Organ of corti is present inside the : COMP 2021

- (a) Cochlea
- (b) Semicircular canals
- (c) Sacculus
- (d) Utriculus

15. The region where the image is sharpest is known as MAIN 2023

- (a) blind spot
- (b) rod region
- (c) fovea
- (d) retina

16. Cerebrum is a part of COMP 2023

- (a) forebrain
- (b) mid brain
- (c) neuron

(d) hind brain

17. A biology student is taught that the nervous system is made up of specialized cells responsible for transmitting signals throughout the body. Which of the following is the functional unit of the nervous system?

MAIN2023

(a) Cyton

(b) Perikaryon

(c) Axon

(d) Neuron

18. Impulses travel much more rapidly along myelinated nerve. SQP 2022

19. The tissue that connects two cerebral hemispheres. SQP 2022

20. The nerve that carries impulses from brain to various tissues. SQP 2023

21. A mixed nerve MAIN 2022

(a) carries sensations from two or more different sense organs

(b) has a common root but branches into two nerves to different organs

(c) contains both sensory and motor fibres

(d) has two or more roots from different parts of brain

22. A synapse is found between MAIN 2022

(a) axon terminal and axon terminal

(b) dendrite and axon terminal

(c) dendrite and dendrite

(d) All of the above

23. Name the nerve that transmits impulses to the brain from : SQP 2021

(a) Ear

(b) Eye

24. Auditory canal, Organ of Corti, Stapes, Malleus. SQP 2025

25. A student visits the eye doctor with sensitivity to bright light. The doctor explains that the circular opening in the center of the iris controls the amount of light entering the eye. The circular opening in the centre of iris : SQP 2021

(a) Lens

(b) Cornea

(c) Sclera

(d) Pupil

26. The fluid in which the membranous labyrinth of inner ear floats is perilymph. COMP 2023

27. In presbyopia loss of elasticity in the lens occurs with ageing. MAIN 2021

28. The muscles which help in alteration of shape of the lens. SQP 2023

29. The nerves which transmit impulses from ear to the brain. SQP 2021

30. Name the three membranous coverings of the human brain. MAIN 2021

31. ____receive information from sensory receptors and transmit it to the central nervous system. COMP 2021

32. The cavities of brain are called____. MAIN 2022

33. What are eyelids ? SQP 2022

34. Define the term Synapse. How are Cytons and Axons of neurons placed in the Cerebrum ? COMP 2021

35. Applying brakes of a car SQP 2022

36. Knitting without looking MAIN 2023

37. What is organ of corti ? SQP 2022

38. Chris was watching the display of fireworks in the sky.

SQP 2025



(a) Trace the path of the light rays using the following terms: Fovea, Lens, Conjunctiva, Pupil, Cornea

(b) Name the nerve that carries the impulse for vision to the brain.

39. Which parts of the ear are responsible for

MAIN 2024

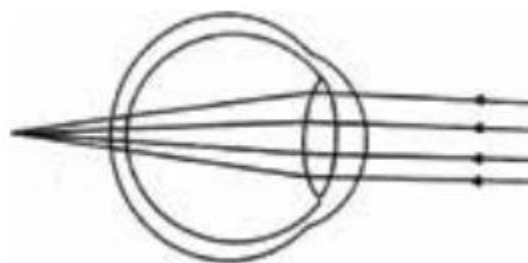
(i) Static equilibrium?

(ii) Dynamic equilibrium?

40. Draw a neat diagram of a Neuron and label any two parts.

COMP 2021

41. Study the diagram given below and answer the questions that follow :



Sem II, 2021

(a) Identify the defect of the eye by mentioning the technical term.

(b) Mention one reason for this defect.

(c) Name the type of lens used to correct this defect.

Solution

1. (c) Central nervous system

Reflex action is controlled by the **central nervous system**, mainly by the **spinal cord**.

2. (c) Lysozyme

3. (d) Corpus callosum

4. (a) Rupture of eardrum

5. (b) Assertion is False and Reason is True.

6. (c) Ciliary outgrips

7. (a) Stapes

8. Spinal cord

9. **Pinna** – It receives sound vibrations from the surroundings and directs them into the **auditory canal**.

10. **Eardrum** – It receives sound vibrations from the auditory canal and transmits them to the ear ossicles.

11. **Eustachian tube** – It helps in **equalizing the air pressure** on both sides of the **eardrum**.

12. The two hemispheres of the cerebrum are connected by the **corpus callosum**.

13. (a) Neurilemma

The insulating sheath covering the axon is called the **neurilemma**.

14. (a) Cochlea

The **organ of Corti** is located inside the **cochlea** of the inner ear. It contains sensory cells responsible for hearing.

15. (c) Fovea

The region where the image is sharpest is called the **fovea**. It has a high concentration of cone cells responsible for clear and detailed vision.

16. (a) Forebrain

The **cerebrum** is a part of the **forebrain**, which is the largest part of the brain. It receives impulses from sensory organs and controls voluntary actions.

17. (d) Neuron

The functional unit of the nervous system is the **neuron**. It receives, processes, and transmits information through electrical and chemical signals.

18. True

19. Corpus callosum

20. Efferent nerve.

21. (c) Contains both sensory and motor fibres

A **mixed nerve** contains both **sensory and motor fibres**, which means it can carry impulses to and from the brain or spinal cord.

22. (b) Dendrite and axon terminal

A **synapse** is found between a **dendrite and an axon terminal**. It allows the transmission of nerve impulses from one neuron to another.

23. (a) From the ear – **Auditory nerve**

(b) From the eye – **Optic nerve**

24. The correct sequence is — Auditory canal → Malleus → Stapes → Organ of Corti

25. (d) Pupil

The **circular opening in the center of the iris** is called the **pupil**. It controls the amount of light entering the eye by adjusting its size.

26. True

27. True

28. Ciliary muscles

29. Auditory nerve

30. Piamater, Arachnoid, Duramater

31. Sensory neurons

32. Ventricles

33. **Eyelids** are two movable protective folds that cover the front (anterior) surface of the eyeball when closed.

34. **Synapse**: It is the point of contact between the terminal branches of the axon of one neuron and the dendrites of another neuron, separated by a fine gap.

Placement in the Cerebrum: The **outer part** of the cerebrum consists of **cytons** (cell bodies), while the **inner part** contains **axons**.

35. Conditioned

36. Conditioned

37. The **Organ of Corti** is an elongated, spiral structure that runs along the entire length of the **cochlea** in the inner ear. It contains sensory cells responsible for hearing.

38. (a) The path of light rays when Chris sees the fireworks:

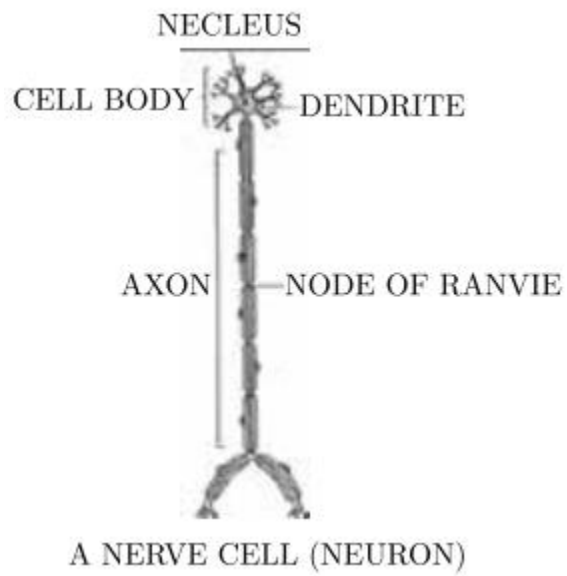
Conjunctiva → Cornea → Pupil → Lens → Fovea

(b) The nerve that carries the visual impulse to the brain: **Optic nerve**.

39. (i) **Static equilibrium** – Maintained by the **utricle** and **sacculle**, which are parts of the **vestibule** in the inner ear.

(ii) **Dynamic equilibrium** – Maintained by the **semicircular canals**, also located in the **inner ear**.

40.



41. (a) Hypermetropia/Hyperopia

(b) Eyeball shortened from front to back

(c) Convex lens